

Nahid Hasan Sagar

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EDUCATION

Indiana University, College of Arts and Sciences

Master of Science in Statistical Science

Bloomington, IN

Expected May 2027

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II; Statistical Consulting; Research in Statistics

University of Dhaka

Bachelor of Science in Physics

Dhaka, Bangladesh

August 2024

RELEVANT EXPERIENCE

Research Assistant, StatMIND Lab

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Implemented and optimized Bayesian computational routines in C++ using Rcpp and RcppArmadillo for high-dimensional neuroimaging data analysis.
- Reproduced original R package outputs and validated new implementations through unit testing, reproducible benchmarking, and numerical checks.
- Delivered a 4 to 5× speedup in core computations by moving numerically intensive work into compiled C++ code while preserving analytical accuracy.
- Built benchmarking workflows to compare runtime, scalability, and model performance across implementations, improving transparency and reproducibility in research processes.

Teaching Assistant, Department of Statistics

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Support undergraduate statistics courses through discussion sections, problem-solving sessions, and individualized guidance during office hours.
- Grade approximately 200 assignments per cycle and provide detailed feedback on statistical reasoning, computation, and interpretation.
- Communicate complex methods clearly to students with varied backgrounds, strengthening conceptual understanding and quantitative confidence.

Senior Mathematics Instructor

2016 – 2025

Udyash Academic & Admission Care

Dhaka, Bangladesh

- Taught and mentored more than 100,000 students through classroom instruction and national online learning platforms.
- Developed structured lesson plans, practice materials, and individualized academic strategies for students preparing for competitive admissions examinations.
- Translated advanced mathematical concepts into clear, practical explanations and helped students with diverse academic backgrounds build strong analytical problem-solving skills.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using BART

R, Bayesian Modeling

- Conducted research on Bayesian Additive Regression Trees for time-series analysis, evaluating predictive performance and model behavior under a rigorous statistical framework.

Ordinary Least Squares from Scratch

Python, NumPy, Pandas, Matplotlib

- Developed an educational implementation of ordinary least squares regression from first principles, emphasizing matrix-based estimation, numerical validation, and transparent model interpretation.

TECHNICAL SKILLS

Quantitative Methods: Bayesian inference, statistical modeling, time-series analysis, regression, model validation, predictive evaluation, experimental design, and computational statistics

Programming: R, Python, C++, MATLAB, Bash, SQL

Tools and Platforms: Git, GitHub, RStudio, VS Code, LaTeX, Jupyter, NumPy, Pandas, Matplotlib

Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & ACTIVITIES

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Collaborated on the development of a comprehensive online Linear Algebra curriculum during the COVID-19 pandemic.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.